

Network Traffic Analysis Report

Wireshark-Based Network Traffic Capture, Analysis, and Investigation

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Project Type: Network Traffic Analysis & Security Investigation

1. Executive Summary

This report documents a Wireshark-based network traffic capture, analysis, and investigation exercise conducted in a Kali Linux virtual environment. Live browser traffic generated through visits to Kali.org and RedHat.com was captured and analyzed using packet inspection, DNS filtering, protocol hierarchy statistics, and conversation analysis. The report also includes containment recommendations, investigation procedures, and supporting validation tools used by SOC analysts.

2. Scope of Assessment

Assessment scope included Kali Linux VM, Wireshark packet analysis, live packet capture, DNS filtering, protocol hierarchy analysis, conversation statistics review, communication pattern interpretation, and security investigation planning.

3. Assessment Methodology

Methodology followed: environment preparation, live traffic generation, packet capture, display filtering, statistical analysis, investigation, and reporting.

4. Environment Preparation

Tools used: Kali Linux VM, Wireshark, active network interface, browser traffic generation. Environment connectivity and capture visibility were validated before testing.

5. Live Packet Capture Process

Traffic capture was initiated on the active network interface. Browser traffic was generated by visiting <https://www.kali.org/> and <https://www.redhat.com/en>. Captured traffic included DNS, TCP, TLS, and HTTPS communications. Capture file saved as Packet capture.pcapng.

6. DNS Traffic Analysis

Display filter used: dns. Analysis identified hostname resolution requests and responses occurring prior to encrypted sessions. DNS monitoring is valuable for detecting malicious domains, beaconing, tunneling, and suspicious resolution behavior.

7. Protocol Hierarchy Analysis

Protocol hierarchy review showed common browsing protocols including IPv4, TCP, TLS/SSL, DNS, and HTTPS. Encrypted traffic represented a significant portion of packet activity.

8. Conversation Statistics Analysis

Conversation analysis reviewed source/destination endpoints, session relationships, and traffic volume sorted by bytes. Traffic behavior aligned with expected client-server browsing activity.

9. Security Findings & Risk Analysis

Observed traffic reflected normal browsing behavior. Potential risk indicators in production environments include suspicious DNS domains, unknown outbound connections, abnormal ports, large transfers, or persistent encrypted tunnels.

10. Immediate Containment Recommendations

If suspicious activity is detected: isolate affected hosts, block malicious IPs/domains, restrict outbound communications, preserve packet captures and logs, and enforce IDS/IPS or firewall controls.

11. Investigation Procedures

Review packet metadata, inspect DNS activity, validate destination reputation, correlate events with SIEM/firewall/endpoint logs, and inspect affected systems for malware indicators or persistence.

12. Supporting Tools & Logs Validation

Validation sources include Wireshark, Zeek, NetFlow analyzers, Snort/Suricata, firewall logs, DNS logs, proxy logs, endpoint telemetry, and SIEM platforms such as Splunk, ELK, or Microsoft Sentinel.

13. Skills Demonstrated

Network Traffic Analysis, Packet Capture, Wireshark Operations, DNS Investigation, Protocol Analysis, Security Monitoring, Incident Investigation, Documentation and Reporting.

14. Conclusion

The exercise demonstrates practical SOC and network security analysis workflows involving packet capture, DNS filtering, protocol investigation, traffic interpretation, and incident response thinking.

Appendix A — Filters and Analysis Steps

Display Filter: dns

Navigation: Statistics → Protocol Hierarchy; Statistics → Conversations

Websites: kali.org, redhat.com

Capture File: Packet capture.pcapng